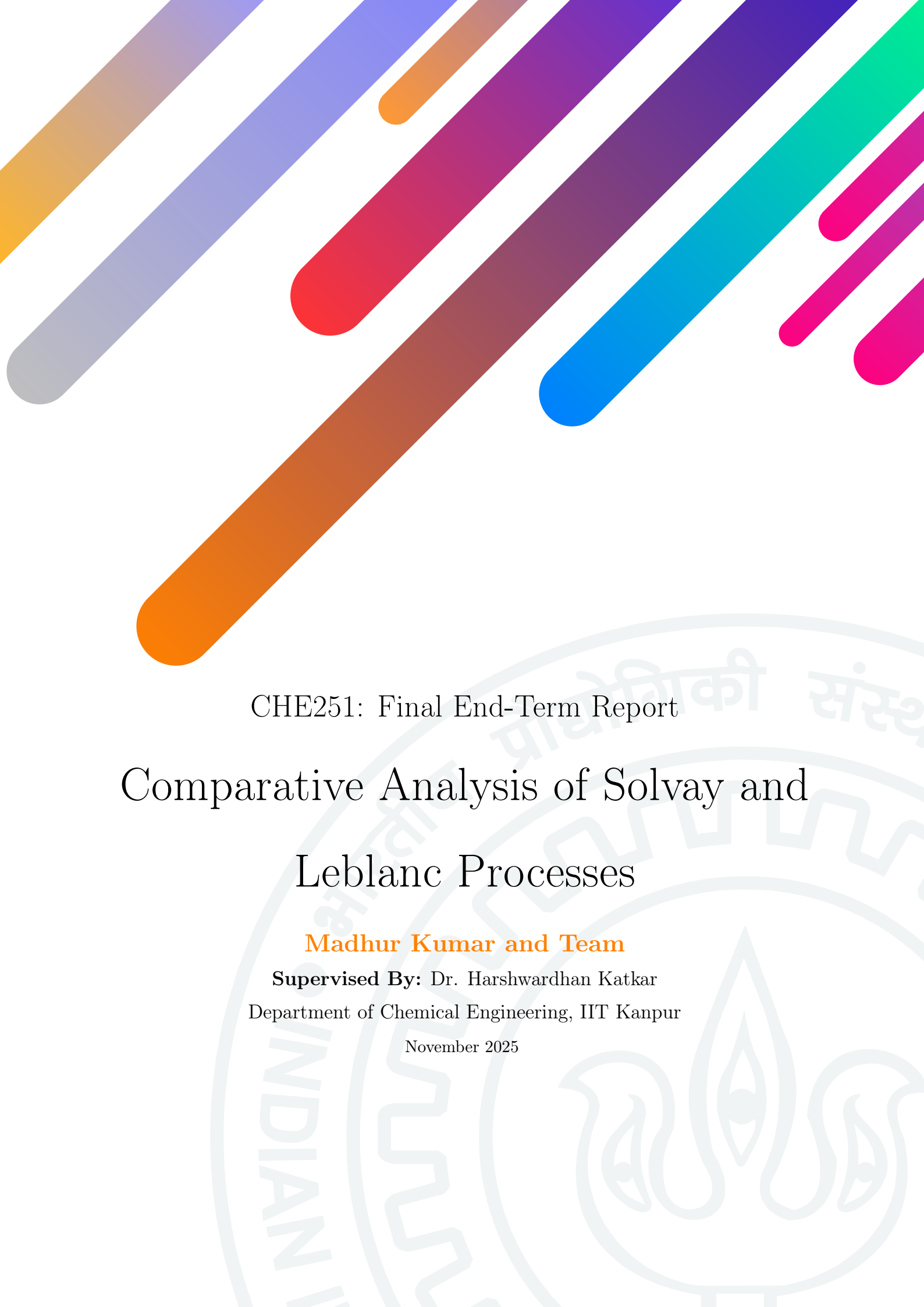




CHE251: Final End-Term Report

Comparative Analysis of Solvay and Leblanc Processes

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1.1 Overview

This chapter describes the numerical and analytical approach used to obtain steady-state material and energy balances for the Solvay and Leblanc process flow sheets. The workflow is:

1. Translate the process flow diagrams and reaction stoichiometry into algebraic mass-balance equations (total and component balances) and reaction extent definitions.
2. Close recycle loops by writing recycle-closure equations (e.g., ammonia recycle, CO_2 recycle).
3. Provide specifications for basis, feed purities and single-pass conversions (these are model inputs).
4. Solve the non-linear algebraic system using MATLAB (root-finding) for stream molar flows and compositions.
5. Compute energy duties using standard enthalpy expressions written relative to 25°C ($C_p dT$ integrals + standard reaction enthalpies).
6. Validate results against industry benchmark values and perform sensitivity analyses.

1.2 Processes in Focus: Solvay and Leblanc

The two industrial processes under study—Solvay and Leblanc—are both designed for the production of sodium carbonate (soda ash, Na_2CO_3), an essential chemical used in glass, detergents, and water treatment.

- The **Solvay Process** is a modern cyclic process that utilizes brine (NaCl), limestone (CaCO_3), and ammonia (NH_3) to produce soda ash while continuously recycling ammonia. It operates with significantly lower waste generation and energy consumption compared to earlier methods.
- The **Leblanc Process** was the first large-scale industrial method for producing Na_2CO_3 , involving the conversion of NaCl to Na_2SO_4 , followed by reduction with carbon and limestone. Although historically significant, it suffered from poor thermal efficiency and large-scale production of calcium sulfide (CaS) waste.

The objective of this study is to compare these processes on the basis of their energy requirements and efficiency. Material and energy balances are computed for both flowsheets using MATLAB, after which energy consumption and thermal efficiency are compared quantitatively to determine the more energy-efficient process.

MATLAB Model Implementation

2.1 Structure

The MATLAB model includes:

- **main.m**: Driver script (loads configuration, runs solver, saves results)
- **mass_balance_system.m**: Material-balance residuals
- **energy_balance.m**: Unit energy duties
- **thermo.m**: Cp and enthalpy data
- **postprocess.m**: Result formatting, plotting

2.2 Solver snippet

```

1 clear; clc;
2 cfg = load('config.mat'); % basis, conversions, temps
3 x0 = initial_guess(cfg);
4
5 opts = optimoptions('fsolve','Display','iter','FunctionTolerance',1e-9);
6 fun = @(x) mass_balance_system(x,cfg);
7 [x_sol,fval,exitflag] = fsolve(fun,x0,opts);
8
9 if exitflag <= 0
10     warning('Solver did not converge');
11 end
12
13 streams = postprocess_solution(x_sol,cfg);
14 energy = energy_balance(streams,cfg);
15 save('results.mat','streams','energy');
```

Listing 2.1: Solver Snippet

2.3 Leblanc Process MATLAB Implementation

The following MATLAB script defines and solves the system of nonlinear equations for the Leblanc process using `lsqnonlin`. The code initializes all flow and composition variables, applies logical bounds, and solves the system of 50 equations (including energy balances) defined in the function `leblanc.m`.

Main Script: leblanc.m

```

1 clear all; clc;
2
3 % INITIALIZATION AND FIXED VALUES
4 F16 = 1000; % Basis: Product Na2CO3 = 1000 kmol/h
5 x1 = 2/3;
6 x5 = 1/3;
7
8 % Initial guesses
9 nVar = 50;
10 x0 = ones(1,nVar) * 100;
11 x0(16) = 1000; % Fix product flow
12 x0(25) = 2/3; x0(29) = 1/3;
13
```

```

14 % Lower and upper bounds
15 lb = zeros(50,1); ub = Inf(50,1);
16 lb(16) = 1000; ub(16) = 1000; % Fix Na2CO3 product
17 lb(25) = 2/3; ub(25) = 2/3;
18 lb(29) = 1/3; ub(29) = 1/3;
19 ub(26:28) = 1.0; ub(30:36) = 1.0;
20 ub(37:42) = 1.0; ub(43:47) = 1.0;
21
22 % Solver options
23 options = optimoptions('lsqnonlin','Display','off','MaxIterations',1000,...
24     'MaxFunctionEvaluations',50000);
25
26 % SOLVE MASS BALANCE
27 x = lsqnonlin(@leblanc,x0,lb,ub,options);
28
29 fprintf('\n LEBLANC PROCESS RESULTS\n');
30 for i=1:length(x)
31     fprintf('x(%02d) = %12.6f\n', i, x(i));
32 end
33
34 % Extract key process quantities
35 F_NaCl = x(1); % Feed NaCl flow (kmol/h)
36 F_Na2CO3 = x(16); % Product Na2CO3 flow (fixed at 1000 kmol/h)
37 ratio = F_NaCl / F_Na2CO3;
38
39 fprintf('Calculated NaCl feed required for %.0f kmol/h Na2CO3:\n', F_Na2CO3);
40 fprintf('NaCl feed (F1): %10.3f kmol/h\n', F_NaCl);
41 fprintf('NaCl to Na2CO3 ratio: %10.3f kmol NaCl per kmol Na2CO3\n', ratio);
42
43
44 % ENERGY BALANCE POST-CALCULATION
45 fprintf('\n ENERGY BALANCE RESULTS\n');
46
47 % Basis and conversions
48 basis = F_Na2CO3; % kmol/h Na2CO3 (product)
49 mol_per_h = basis * 1000; % mol/h
50 M_Na2CO3 = 105.99; % g/mol
51 t_per_h = basis * M_Na2CO3 / 1000; % tonnes/h of Na2CO3
52
53 % Extract reaction extents from solver
54 e1_kmol_h = x(48);
55 e2_kmol_h = x(49);
56 e3_kmol_h = x(50);
57 e1_mol_h = e1_kmol_h * 1000;
58 e2_mol_h = e2_kmol_h * 1000;
59 e3_mol_h = e3_kmol_h * 1000;
60
61 % Cp values (J/mol*K)
62 Cp = struct('NaCl',50,'H2SO4',84,'Na2SO4',100,'HCl',29,...
63     'CaCO3',82,'CaO',50,'CO2',37,'Na2CO3',112,'H2O',34,'CaS',50);
64
65 % Hr (kJ/mol)
66 dH = struct('reactor1',-160,... % 2NaCl + H2SO4 Na2SO4 + 2HCl
67     'reactor2', 700,... % Na2SO4 + CaCO3 + 2C Na2CO3 + CaS + 2CO2
68     'leach', 20); % Na2CO3 + H2O Na2CO3(aq)
69
70 % Convert Hr to J/mol
71 dH = structfun(@(x) x*1e3, dH, 'UniformOutput', false);
72
73 % Temperatures ( C )
74 T = struct('reactor1',400,'reactor2',1000,'leach',60);
75
76 % Kirchhoff

```

```

77 corr_dH = @(dH25,nu,CpVals,T) dH25 + sum(nu .* CpVals) * (T - 25);
78
79 % 1. REACTOR I - 2NaCl + H2SO4      Na2SO4 + 2HCl
80 nu_R1 = [2 1 -1 -2]; % NaCl, H2SO4, Na2SO4, HCl
81 Cp_R1 = [Cp.NaCl Cp.H2SO4 Cp.Na2SO4 Cp.HCl];
82 dH_R1 = corr_dH(dH.reactor1, nu_R1, Cp_R1, T.reactor1);
83 extent_R1 = e1_mol_h;
84 Q_R1 = extent_R1 * dH_R1;
85
86 % 2. REACTOR II      Na2SO4 + CaCO3 + 2C      Na2CO3 + CaS + 2CO2
87 nu_R2 = [1 1 2 -1 -1 -2]; % Na2SO4 CaCO3 C Na2CO3 CaS CO2
88 Cp_R2 = [Cp.Na2SO4 Cp.CaCO3 8 Cp.Na2CO3 Cp.CaS Cp.CO2];
89 dH_R2 = corr_dH(dH.reactor2, nu_R2, Cp_R2, T.reactor2);
90 extent_R2 = e2_mol_h;
91 Q_R2 = extent_R2 * dH_R2;
92
93 % 3. LEACHING      Na2CO3 + H2O      Na2CO3(aq)
94 nu_R3 = [1 1 -1];
95 Cp_R3 = [Cp.Na2CO3 Cp.H2O Cp.Na2CO3];
96 dH_R3 = corr_dH(dH.leach, nu_R3, Cp_R3, T.leach);
97 extent_R3 = e3_mol_h;
98 Q_R3 = extent_R3 * dH_R3;
99
100 % 4. TOTAL ENERGY DUTY
101 Q_total = Q_R1 + Q_R2 + Q_R3;
102 Q_total_GJh = Q_total / 1e9;
103 energy_per_tonne = Q_total_GJh / t_per_h;
104
105 % DISPLAY ENERGY SUMMARY
106
107 fprintf('Unit-wise energy duties (GJ/h):\n');
108 fprintf('Reactor I (400 C ):      %+10.4f\n', Q_R1/1e9);
109 fprintf('Reactor II (1000 C ):      %+10.4f\n', Q_R2/1e9);
110 fprintf('Leaching (60 C ):          %+10.4f\n', Q_R3/1e9);
111 fprintf('-----\n');
112 fprintf('Total energy duty (GJ/h): %+10.4f\n', Q_total_GJh);
113 fprintf('Energy per tonne Na2CO3 : %+10.4f GJ/t\n', energy_per_tonne);

```

Listing 2.2: Main script for solving the Leblanc process system

Function Definition: leblanc.m

```

1 function L = leblanc(x)
2
3     F = x(1); F2 = x(2); F3 = x(3); F4 = x(4); F5 = x(5); F6 = x(6);
4     F7 = x(7); F8 = x(8); F9 = x(9); F10 = x(10); F11 = x(11);
5     F12 = x(12); F13 = x(13); F14 = x(14); F15 = x(15); F16 = x(16);
6     F17 = x(17); F18 = x(18); F19 = x(19); F20 = x(20); F21 = x(21);
7     F22 = x(22); F23 = x(23); F24 = x(24);
8     x1 = x(25); x2 = x(26); x3 = x(27); x4 = x(28); x5 = x(29);
9     x6 = x(30); x7 = x(31); x8 = x(32); x9 = x(33); x10 = x(34);
10    x11 = x(35); x12 = x(36);
11    y3 = x(37); y6 = x(38); y7 = x(39); y9 = x(40); y11 = x(41);
12    y12 = x(42);
13    z3 = x(43); z6 = x(44); z9 = x(45);
14    w6 = x(46); v6 = x(47);
15    e1 = x(48); e2 = x(49); e3 = x(50);
16
17    % mass-balance equations
18    L = zeros(50,1);
19    L(1) = 2*F16 - F*x1; % Determine F from Na2CO3 output
20    L(2) = x1 - 2*(1-x1);
21    L(3) = F3*x3 - F2*x2 + 2*e1;

```

```

22 L(4) = F3*y3 - F2*(1-x2) + e1;
23 L(5) = F3*z3 - e1;
24 L(6) = F3*(1-x3-y3-z3) - 2*e1;
25 L(7) = 2*e1 - F2*x2*0.98;
26 L(8) = F + F7 - F2;
27 L(9) = F*(x1) + F7*x4 - F2*x2;
28 L(10) = F3 - F4 - F8 - F5;
29 L(11) = F3*x3 - F5*x4;
30 L(12) = F3*y3 - F5*(1-x4);
31 L(13) = F3*z3 - F8;
32 L(14) = F5 - F7 - F6;
33 L(15) = F*x1 - 2*F9*x5;
34 L(16) = 2*x5 - (1-x5);
35 L(17) = F11*x6 - F24*y11 + e2;
36 L(18) = F11*y6 - F24*x11 + e2;
37 L(19) = F11*z6 - F24*(1-x11-y11) + 2*e2;
38 L(20) = F11*w6 - e2;
39 L(21) = F11*v6 - e2;
40 L(22) = F11*(1-x6-y6-z6-w6-v6) - 2*e2;
41 L(23) = e2 - 0.9*F24*y11;
42 L(24) = F8 + F9 + F10 - F24;
43 L(25) = F8 + F10*y7 - F24*y11;
44 L(26) = F9*x5 + F10*x7 - F24*x11;
45 L(27) = F11 - F12 - F13 - F15;
46 L(28) = F11*x6 - F12*y7;
47 L(29) = F11*y6 - F12*x7;
48 L(30) = F11*z6 - F12*(1-x7-y7);
49 L(31) = F11*w6 - F15*x8;
50 L(32) = F11*v6 - F15*(1-x8);
51 L(33) = F12 - F10 - F14;
52 L(34) = 0; % Replaced with relation in L(1)
53 L(35) = F18*y9 - F23*x12 + e3;
54 L(36) = F18*x9 - e3;
55 L(37) = F18*z9 - F23*(1-x12-y12) + 12.5*e3;
56 L(38) = F18*(1-x9-y9-z9) - F23*y12;
57 L(39) = F15 + F16 + F17 - F23;
58 L(40) = F15*x8 + F17*x10 - F23*x12;
59 L(41) = F15*(1-x8) - F23*y12;
60 L(42) = e3 - 0.95*F23*x12;
61 L(43) = F18 - F19 - F20 - F21;
62 L(44) = F18*x9 - F20;
63 L(45) = F18*y9 - F19*x10;
64 L(46) = F18*z9 - F19*(1-x10);
65 L(47) = F19 - F17 - F22;
66 end

```

Listing 2.3: Main script for solving the Leblanc process system.

This script employs `lsqnonlin` to minimize the residual vector L for the system of 50 algebraic equations. The computed variable vector corresponds to the steady-state molar flows, compositions, and reaction extents for each unit operation of the Leblanc process.

2.4 Solvay Process MATLAB Implementation

The following MATLAB script defines and solves the system of nonlinear algebraic equations representing the steady-state material balance of the Solvay Process for sodium carbonate production. Each equation corresponds to a unit operation shown in the process flow diagram, including the kiln, lime dissolver, absorber, carbonation tower (Solvay tower), light ash calciner, distiller, and the final recycle loops.

The model uses `fsolve()` to determine the set of flow rates, compositions, and extents of reactions that satisfy the overall and component mass balances for the process. The script

assumes a basis of 1000 kmol/h of sodium carbonate production and defines all stream variables (F2, F3, ..., F34) and component mole fractions (x, y, z, u, w) as symbolic unknowns.

Main Script: solvay.m

```

1 clear all; clc;
2
3 %initial guesses
4 F2=100; F3=100; F4=100; F5=100; F6=100; F7=100; F8=100; F9=100; F10=100;
5 F11=100; F13=100; F14=100; F15=100; F16=100; F17=100; F18=100; F19=100;
6 F20=100; F21=100; F22=100; F23=100; F24=100; F25=100; F26=100; F27=100;
7 F28=100; F29=100; F30=100; F31=100; F32=100; F33=100; F34=100;
8
9 x1=0.5; x2=0.3; x3=0.3; x4=0.4; x5=0.3; x6=0.3; x7=0.4; x8=0.4;
10 x9=0.3; x10=0.4; x11=0.4; x13=0.4;
11 y2=0.2; y5=0.3; y6=0.3; y9=0.2; y11=0.3;
12 z2=0.2; z6=0.2; z9=0.2; z11=0.2;
13 u6=0.1; u9=0.1; w6=0.1;
14 e1=10; e2=10;
15
16 x0 = [F2,F3,F4,F5,F6,F7,F8,F9,F10,F11,F13,F14,F15,F16,F17,F18,F19,F20,F21,F22,F23,F24,F25,F26,
17       x1,x2,x3,x4,x5,x6,x7,x8,x9,x10,x11,x13,y2,y5,y6,y9,y11,z2,z6,z9,z11,u6,u9,w6,e1,e2];
18
19 opts = optimoptions('fsolve','Display','off','FunctionTolerance',1e-9,'MaxFunctionEvaluations',1000);
20 [x, fval, exitflag] = fsolve(@solvay,x0,opts);
21
22 fprintf('\n===== MATERIAL BALANCE RESULTS =====\n');
23 var_names = ["F2","F3","F4","F5","F6","F7","F8","F9","F10","F11","F13","F14","F15","F16","F17",
24             "x1","x2","x3","x4","x5","x6","x7","x8","x9","x10","x11","x13","y2","y5","y6","y9","y11","z2","z6","z9","z11","u6","u9","w6","e1","e2"];
25 for i = 1:length(x)
26     fprintf('%-4s = %12.6f\n', var_names(i), x(i));
27 end
28
29 % ENERGY BALANCE SECTION (BASIS = Na2CO3 = 1000 mol/h)
30
31 fprintf('\n===== ENERGY BALANCE RESULTS =====\n');
32
33 % ---- BASIS ----
34 basis_Na2CO3_mol_per_h = 1000; % basis: Na2CO3 flow = 1000 mol/h
35 mol_per_h = basis_Na2CO3_mol_per_h; % mol/h (for per-mol energy calc)
36 M_Na2CO3 = 105.99; % g/mol (molar mass Na2CO3)
37 % convert to tonnes per hour: (mol/h * g/mol) / 1e6 = tonnes/h
38 t_per_h = mol_per_h * M_Na2CO3 / 1e6;
39
40 fprintf('Basis: Na2CO3 = %g mol/h (%.6f t/h)\n', basis_Na2CO3_mol_per_h, t_per_h);
41
42 % Cp (J/mol K)
43 Cp = struct('CaCO3',82,'CaO',50,'CO2',37,'NaHCO3',102,'Na2CO3',112,...
44            'H2O',34,'NH3',35,'NaCl',50,'NH4Cl',70);
45
46 % Hr at 25 C (kJ/mol)
47 dH25 = struct(...
48     'kiln', +178, ... % CaCO3 CaO + CO2
49     'lime', -64, ... % CaO + H2O Ca(OH)2
50     'NH3abs', -35, ... % NH3 + H2O NH4OH
51     'CO2abs', -45, ... % CO2 + NH3 + H2O NH4HCO3
52     'solvay', -85, ... % NaCl + NH3 + CO2 + H2O NaHCO3 + NH4Cl
53     'filter', -10, ... % NaHCO3 crystallization
54     'calciner', +135, ... % 2 NaHCO3 Na2CO3 + H2O + CO2
55     'monohyd', -41, ... % Na2CO3 + H2O Na2CO3 H2O
56     'distiller', +75, ... % 2 NH4Cl + Ca(OH)2 2 NH3 + CaCl2 + 2 H2O
57     'compressor', +10, ... % CO2 compression (work)
58 );
59

```



```

60
61 % Convert to J/mol
62 fields = fieldnames(dH25);
63 for k = 1:length(fields)
64     dH25.(fields{k}) = dH25.(fields{k}) * 1e3;
65 end
66
67 % Temperatures ( C )
68 T = struct('kiln',1000,'lime',100,'NH3abs',30,'CO2abs',40,'tower',50,...
69           'filter',25,'calciner',200,'monohyd',40,'distiller',100,'compressor',50);
70
71 % Kirchhoff correction (linear Cp approximation)
72 % dH(T) = dH(25) + sum( nu_i * Cp_i ) * (T - 25)
73 corr_dH = @(dH0, nu_vec, Cp_vec, Tunit) dH0 + sum(nu_vec .* Cp_vec) * (Tunit - 25);
74
75
76 % Individual units (use mol_per_h basis)
77
78 % Kiln: CaCO3 -> CaO + CO2
79 nu_kiln = [-1 +1 +1];
80 Cp_kiln = [Cp.CaCO3, Cp.CaO, Cp.CO2];
81 Q_kiln = mol_per_h * corr_dH(dH25.kiln, nu_kiln, Cp_kiln, T.kiln);
82
83 % Lime dissolver: CaO + H2O -> Ca(OH)2 (example Cp for Ca(OH)2 set to 90 J/mol K)
84 nu_lime = [-1 -1 +1];
85 Cp_lime = [Cp.CaO, Cp.H2O, 90];
86 Q_lime = mol_per_h * corr_dH(dH25.lime, nu_lime, Cp_lime, T.lime);
87
88 % NH3 absorber: NH3 + H2O -> NH4OH (example Cp for NH4OH set to 70 J/mol K)
89 nu_NH3 = [-1 -1 +1];
90 Cp_NH3 = [Cp.NH3, Cp.H2O, 70];
91 Q_NH3 = mol_per_h * corr_dH(dH25.NH3abs, nu_NH3, Cp_NH3, T.NH3abs);
92
93 % CO2 absorber: CO2 + NH3 + H2O -> NH4HCO3
94 nu_CO2 = [-1 -1 -1 +1];
95 Cp_CO2 = [Cp.CO2, Cp.NH3, Cp.H2O, Cp.NaHCO3];
96 Q_CO2 = mol_per_h * corr_dH(dH25.CO2abs, nu_CO2, Cp_CO2, T.CO2abs);
97
98 % Solvay tower (extent assumed 0.9 * mol_per_h)
99 nu_tower = [-1 -1 -1 -1 +1 +1];
100 Cp_tower = [Cp.NaCl, Cp.NH3, Cp.CO2, Cp.H2O, Cp.NaHCO3, Cp.NH4Cl];
101 extent_tower = 0.9 * mol_per_h;
102 Q_tower = extent_tower * corr_dH(dH25.solvay, nu_tower, Cp_tower, T.tower);
103
104 % Filter / crystallizer
105 nu_filter = [-1 +1];
106 Cp_filter = [Cp.NaHCO3, Cp.NaHCO3];
107 Q_filter = mol_per_h * corr_dH(dH25.filter, nu_filter, Cp_filter, T.filter);
108
109 % Calciner: 2 NaHCO3 -> Na2CO3 + H2O + CO2
110 nu_calc = [-2 +1 +1 +1];
111 Cp_calc = [Cp.NaHCO3, Cp.Na2CO3, Cp.H2O, Cp.CO2];
112 Q_calc = mol_per_h * corr_dH(dH25.calciner, nu_calc, Cp_calc, T.calciner);
113
114 % Monohydrate formation: Na2CO3 + H2O -> Na2CO3 H2O (example Cp for hydrate = 130)
115 nu_mon = [-1 -1 +1];
116 Cp_mon = [Cp.Na2CO3, Cp.H2O, 130];
117 Q_mon = mol_per_h * corr_dH(dH25.monohyd, nu_mon, Cp_mon, T.monohyd);
118
119 % Distiller / ammonia recovery
120 nu_dist = [-2 -1 +2 +1 +2];
121 Cp_dist = [Cp.NH4Cl, 90, Cp.NH3, 70, Cp.H2O]; % some example Cp placeholders
122 Q_dist = mol_per_h * corr_dH(dH25.distiller, nu_dist, Cp_dist, T.distiller);

```

```

123
124 % CO2 compressor
125 nu_comp = [1];
126 Cp_comp = [Cp.CO2];
127 Q_comp = mol_per_h * corr_dH(dH25.compressor, nu_comp, Cp_comp, T.compressor);
128
129
130 Q_units = [Q_kiln, Q_lime, Q_NH3, Q_CO2, Q_tower, Q_filter, Q_calc, Q_mon, Q_dist, Q_comp];
131 names = {'Kiln', 'Lime Dissolver', 'NH3 Absorber', 'CO2 Absorber', 'Solvay Tower', ...
132         'Filter/Crystallizer', 'Calciner', 'Monohydrate Mixer', 'Distiller', 'CO2 Compressor'};
133
134 fprintf('\nEnergy Duties by Unit (GJ/h):\n');
135 for i=1:length(Q_units)
136     fprintf('%-22s : %+10.6f\n', names{i}, Q_units(i)/1e9);
137 end
138
139 Q_total = sum(Q_units);
140 Q_total_GJh = Q_total/1e9;
141 energy_per_tonne = Q_total_GJh / t_per_h;
142
143 % Heat recovery (typical 70% of exothermic heat reused)
144 Q_exo = sum(Q_units(Q_units<0));
145 Q_recovery = 0.7 * abs(Q_exo);
146 Q_net = Q_total + Q_recovery; % recovered heat reduces net input
147 Q_net_GJh = Q_net / 1e9;
148 energy_net_tonne = Q_net_GJh / t_per_h;
149
150 fprintf('-----\n');
151 fprintf('Total (gross) energy duty (GJ/h): %+10.6f\n', Q_total_GJh);
152 fprintf('Energy per tonne Na2CO3 (gross): %+10.6f GJ/t\n', energy_per_tonne);
153 fprintf('Recovered heat (70%% exotherm): %+10.6f GJ/h\n', Q_recovery/1e9);
154 fprintf('Net energy after recovery (GJ/h): %+10.6f\n', Q_net_GJh);
155 fprintf('Net energy per tonne Na2CO3: %+10.6f GJ/t\n', energy_net_tonne);
156 fprintf('-----\n');
157
158
159 % MATERIAL BALANCE FUNCTION
160
161 function S = solvay(x)
162
163 F2 = x(1); F3 = x(2); F4 = x(3); F5 = x(4); F6 = x(5); F7 = x(6); F8 = x(7);
164 F9 = x(8); F10 = x(9); F11 = x(10); F13 = x(11); F14 = x(12); F15 = x(13);
165 F16 = x(14); F17 = x(15); F18 = x(16); F19 = x(17); F20 = x(18); F21 = x(19);
166 F22 = x(20); F23 = x(21); F24 = x(22); F25 = x(23); F26 = x(24); F27 = x(25);
167 F28 = x(26); F29 = x(27); F30 = x(28); F31 = x(29); F32 = x(30); F33 = x(31);
168 F34 = x(32); x1 = x(33); x2 = x(34); x3 = x(35); x4 = x(36); x5 = x(37);
169 x6 = x(38); x7 = x(39); x8 = x(40); x9 = x(41); x10 = x(42); x11 = x(43);
170 x13 = x(44); y2 = x(45); y5 = x(46); y6 = x(47); y9 = x(48); y11 = x(49);
171 z2 = x(50); z6 = x(51); z9 = x(52); z11 = x(53); u6 = x(54); u9 = x(55);
172 w6 = x(56); e1 = x(57); e2 = x(58);
173
174 F = 1000;
175 x12 = 0.25; y12 = 0.25; z12 = 0.25;
176
177 S(1,1) = x1 - (1-x1);
178 S(2,1) = 0.21*F2 - F*x1;
179 S(3,1) = F*x1 - F3*x2;
180 S(4,1) = 2*F*x1 - F3*y2;
181 S(5,1) = F3*z2 - 0.21*F2;
182 S(6,1) = F3*(1-x2-y2-z2) - 0.79*F2;
183 S(7,1) = F3*x2 - F27;
184 S(8,1) = F3*y2 - F5;
185 S(9,1) = F3*z2 - F4*x3;

```

```

186 S(10,1) = F3*(1-x2-y2-z2) - F4*(1-x3);
187 S(11,1) = F6 - F27;
188 S(12,1) = F18 - F27;
189 S(13,1) = 3*58.44*x4 - 18.01528*(1-x4);
190 S(14,1) = F17*x4 - F8*x5;
191 S(15,1) = F7*(1-x4) - F8*y5;
192 S(16,1) = F33 - F8*(1-x5-y5);
193 S(17,1) = F5 + F10 - F9;
194 S(18,1) = F28*x13 - e1 - F16*z9;
195 S(19,1) = F28*(1-x13) - 2*e1 - F16*u9;
196 S(20,1) = F16*(1-x9-y9-z9-u9) - 2*e1;
197 S(21,1) = F16*x9 - e1;
198 S(22,1) = F16*y9 - e1;
199 S(23,1) = 0.95*F28*(1-x13) - 2*e1;
200 S(24,1) = F16 - F14 - F29 - F33;
201 S(25,1) = F16*x9 - F14*x7;
202 S(26,1) = F16*y9 - F14*(1-x7);
203 S(27,1) = F16*z9 - F29*x8;
204 S(28,1) = F16*u9 - F29*(1-x8);
205 S(29,1) = F29 - F17 - F30;
206 S(30,1) = F17 + F18 - F28;
207 S(31,1) = F17*x8 + F18 - F28*x13;
208 S(32,1) = F31*z12 - e2 - F11*u6;
209 S(33,1) = F31*y12 - e2 - F11*w6;
210 S(34,1) = F31*(1-x12-y12-z12) - e2 - F11*(1-x6-y6-z6-w6 -u6);
211 S(35,1) = F31*x12 - e2 - F11*z6;
212 S(36,1) = F11*x6 - e2;
213 S(37,1) = F11*y6 - e2;
214 S(38,1) = 0.90*F31*x12 - e2;
215 S(39,1) = F11 - F15 - F32;
216 S(40,1) = F11*x6 - F15*x10;
217 S(41,1) = F11*y6 - F15*(1-x10);
218 S(42,1) = F11*z6 - F32*x11;
219 S(43,1) = F11*w6 - F32*y11;
220 S(44,1) = F11*u6 - F32*z11;
221 S(45,1) = F13 + F32 - F34;
222 S(46,1) = F8 + F13 - F31;
223 S(47,1) = F8*x5 + F13*x11 - F31*x12;
224 S(48,1) = F8*y5 + F13*y11 - F31*y12;
225 S(49,1) = F8*(1-x5-y5) + F13*z11 - F31*z12;
226 S(50,1) = F15 - F19 - F20;
227 S(51,1) = F15*x10 - F19;
228 S(52,1) = F20 - 2*F22;
229 S(53,1) = 2*F21 - F20;
230 S(54,1) = F20 - 2*F10;
231 S(55,1) = F22 - F24;
232 S(56,1) = F23 - 10*F22;
233 S(57,1) = F24 - F26;
234 S(58,1) = F25 - 10*F24;
235 end

```

Listing 2.4: MATLAB code for the Solvay Process System of Equations

The function `solvay(x)` defines the 58 equations of the system, which include:

- Material balances for major components ($NaCl$, NH_3 , CO_2 , $NaHCO_3$, Na_2CO_3 , NH_4Cl , $Ca(OH)_2$).
- Stoichiometric constraints for the primary reactions in the carbonation and recovery sections.
- Recycle and bypass relations connecting major units (e.g., absorber $\rightleftharpoons$ distiller $\rightleftharpoons$ compressor loops).

- Empirical conversion relationships for the ammonia absorber and carbonation tower.
- After initialization with reasonable guesses, the solver iteratively adjusts all unknowns until the residuals of the equations approach zero, signifying convergence to a steady-state solution.

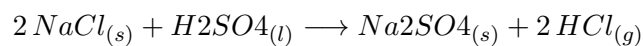
Process Specifications: Leblanc and Solvay

3.1 Leblanc Process: Reactions and Operating Conditions

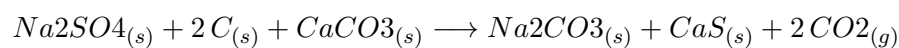
3.1.1 Key Leblanc Process Reactions

The major chemical transformations in the Leblanc process are summarized below. Each reaction corresponds to a distinct unit operation in the process flow.

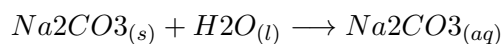
(1) Reactor I — Salt Cake Furnace:



(2) Reactor II — Black Ash Furnace:



(3) Leaching Tank:



3.1.2 Labalanc Flowsheet

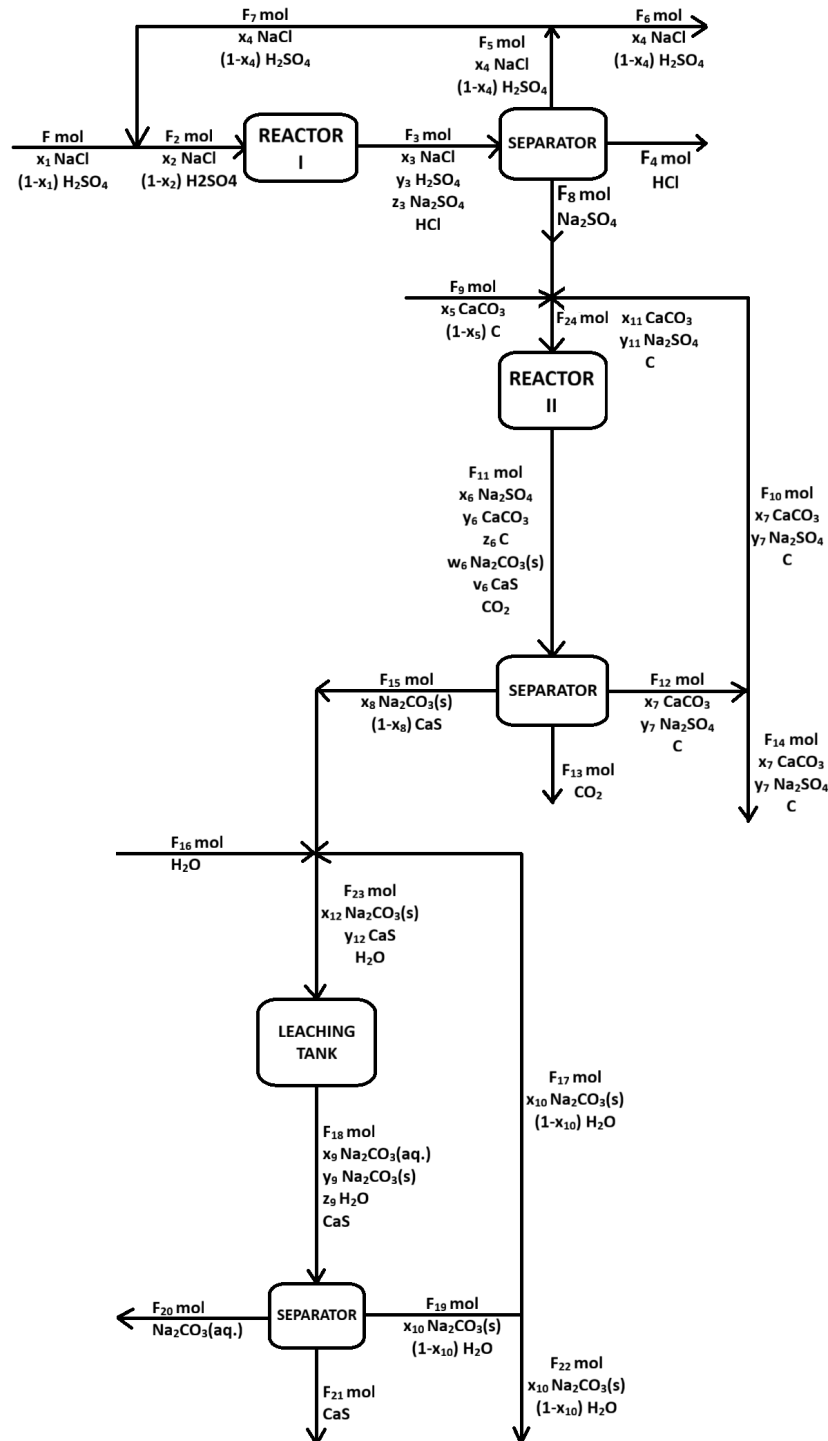


Figure 3.1: Fully Labeled Process Flow Diagram for the Leblanc Process.

3.1.3 Operating Conditions (Leblanc)

Table 3.1: Operating Data for Key Leblanc Unit Operations

Unit Operation	Temperature (T)	Pressure (P)	Conversion / Solubility
Reactor I	600–700°C	1 atm	Single-Pass Conversion (NaCl): ~98%
Reactor II	1000–1100°C	1 atm	Single-Pass Conversion (Na ₂ SO ₄): 85–95%
Leaching Tank	50–70°C	1 atm	Single-Pass Conversion (Na ₂ CO ₃ dissolution): 90–97%

3.1.4 Model Equations for the Leblanc Process

The following system of nonlinear algebraic equations represents the overall and component mass balances for each unit in the Leblanc process. Each equation corresponds to a steady-state constraint in the MATLAB model implemented using `fsolve()`.

Eq. 1. $x_1 - (1 - x_1) = 0$

Eq. 2. $0.21F_2 - Fx_1 = 0$

Eq. 3. $Fx_1 - F_3x_2 = 0$

Eq. 4. $2Fx_1 - F_3y_2 = 0$

Eq. 5. $F_3z_2 - 0.21F_2 = 0$

Eq. 6. $F_3(1 - x_2 - y_2 - z_2) - 0.79F_2 = 0$

Eq. 7. $F_3x_2 - F_{27} = 0$

Eq. 8. $F_3y_2 - F_5 = 0$

Eq. 9. $F_3z_2 - F_4x_3 = 0$

Eq. 10. $F_3(1 - x_2 - y_2 - z_2) - F_4(1 - x_3) = 0$

Eq. 11. $F_6 - F_{27} = 0$

Eq. 12. $F_{18} - F_{27} = 0$

Eq. 13. $3(58.44)x_4 - 18.01528(1 - x_4) = 0$

Eq. 14. $F_{17}x_4 - F_8x_5 = 0$

Eq. 15. $F_7(1 - x_4) - F_8y_5 = 0$

Eq. 16. $F_{33} - F_8(1 - x_5 - y_5) = 0$

Eq. 17. $F_5 + F_{10} - F_9 = 0$

Eq. 18. $F_{28}x_{13} - e_1 - F_{16}z_9 = 0$

Eq. 19. $F_{28}(1 - x_{13}) - 2e_1 - F_{16}u_9 = 0$

Eq. 20. $F_{16}(1 - x_9 - y_9 - z_9 - u_9) - 2e_1 = 0$

Eq. 21. $F_{16}x_9 - e_1 = 0$

Eq. 22. $F_{16}y_9 - e_1 = 0$

Eq. 23. $0.95F_{28}(1 - x_{13}) - 2e_1 = 0$

Eq. 24. $F_{16} - F_{14} - F_{29} - F_{33} = 0$

Eq. 25. $F_{16}x_9 - F_{14}x_7 = 0$

Eq. 26. $F_{16}y_9 - F_{14}(1 - x_7) = 0$

Eq. 27. $F_{16}z_9 - F_{29}x_8 = 0$

Eq. 28. $F_{16}u_9 - F_{29}(1 - x_8) = 0$

Eq. 29. $F_{29} - F_{17} - F_{30} = 0$

Eq. 30. $F_{17} + F_{18} - F_{28} = 0$

Eq. 31. $F_{17}x_8 + F_{18} - F_{28}x_{13} = 0$

Eq. 32. $F_{31}z_{12} - e_2 - F_{11}u_6 = 0$

Eq. 33. $F_{31}y_{12} - e_2 - F_{11}w_6 = 0$

Eq. 34. $F_{31}(1 - x_{12} - y_{12} - z_{12}) - e_2 - F_{11}(1 - x_6 - y_6 - z_6 - w_6 - u_6) = 0$

Eq. 35. $F_{31}x_{12} - e_2 - F_{11}z_6 = 0$

Eq. 36. $F_{11}x_6 - e_2 = 0$

Eq. 37. $F_{11}y_6 - e_2 = 0$

Eq. 38. $0.90F_{31}x_{12} - e_2 = 0$

Eq. 39. $F_{11} - F_{15} - F_{32} = 0$

Eq. 40. $F_{11}x_6 - F_{15}x_{10} = 0$

Eq. 41. $F_{11}y_6 - F_{15}(1 - x_{10}) = 0$

Eq. 42. $F_{11}z_6 - F_{32}x_{11} = 0$

Eq. 43. $F_{11}w_6 - F_{32}y_{11} = 0$

Eq. 44. $F_{11}u_6 - F_{32}z_{11} = 0$

Eq. 45. $F_{13} + F_{32} - F_{34} = 0$

Eq. 46. $F_8 + F_{13} - F_{31} = 0$

Eq. 47. $F_8x_5 + F_{13}x_{11} - F_{31}x_{12} = 0$

Eq. 48. $F_8y_5 + F_{13}y_{11} - F_{31}y_{12} = 0$

Eq. 49. $F_8(1 - x_5 - y_5) + F_{13}z_{11} - F_{31}z_{12} = 0$

Eq. 50. $F_{15} - F_{19} - F_{20} = 0$

Eq. 51. $F_{15}x_{10} - F_{19} = 0$

Eq. 52. $F_{20} - 2F_{22} = 0$

Eq. 53. $2F_{21} - F_{20} = 0$

Eq. 54. $F_{20} - 2F_{10} = 0$

Eq. 55. $F_{22} - F_{24} = 0$

Eq. 56. $F_{23} - 10F_{22} = 0$

Eq. 57. $F_{24} - F_{26} = 0$

Eq. 58. $F_{25} - 10F_{24} = 0$

3.1.5 Solution: Labalanc**Table 3.2:** Solution Vector

Index	Var.	Solution	Index	Var.	Solution
1	$F2$	2.3810×10^3	30	$F32$	5.6188×10^{-12}
2	$F3$	3.8810×10^3	31	$F33$	2.0699×10^{-12}
3	$F4$	2.3810×10^3	32	$F34$	5.6588×10^{-11}
4	$F5$	1000	33	$x1$	0.5000
5	$F6$	500	34	$x2$	0.1288
6	$F7$	1.8978×10^{-12}	35	$x3$	0.2100
7	$F8$	5.2188×10^{-12}	36	$x4$	0.0932
8	$F9$	1.0000×10^3	37	$x5$	0.3102
9	$F10$	6.3211×10^{-12}	38	$x6$	0.4601
10	$F11$	3.0903×10^{-11}	39	$x7$	0.4883
11	$F13$	5.0969×10^{-11}	40	$x8$	1.0000
12	$F14$	1.2216×10^{-12}	41	$x9$	1.0860×10^{-15}
13	$F15$	2.5285×10^{-11}	42	$x10$	0.4814
14	$F16$	500.0000	43	$x11$	0.2233
15	$F17$	1.5324×10^{-11}	44	$x13$	1.0000
16	$F18$	500	45	$y2$	0.2577
17	$F19$	1.2642×10^{-11}	46	$y5$	0.3502
18	$F20$	1.2642×10^{-11}	47	$y6$	0.4503
19	$F21$	6.3211×10^{-12}	48	$y9$	1.0858×10^{-15}
20	$F22$	6.3211×10^{-12}	49	$y11$	0.1598
21	$F23$	6.3211×10^{-11}	50	$z2$	0.1288
22	$F24$	6.3211×10^{-12}	51	$z6$	1.1291×10^{-4}
23	$F25$	6.3211×10^{-11}	52	$z9$	1.0000
24	$F26$	6.3211×10^{-12}	53	$z11$	0.0842
25	$F27$	500	54	$u6$	0.0301
26	$F28$	500.0000	55	$u9$	1.1116×10^{-16}
27	$F29$	500.0000	56	$w6$	0.0214
28	$F30$	500.0000	57	$e1$	6.1081×10^{-13}
29	$F31$	5.6188×10^{-11}	58	$e2$	1.2642×10^{-11}

3.2 Solvay Process: Reactions and Operating Conditions

3.2.1 Solvay Flowsheet

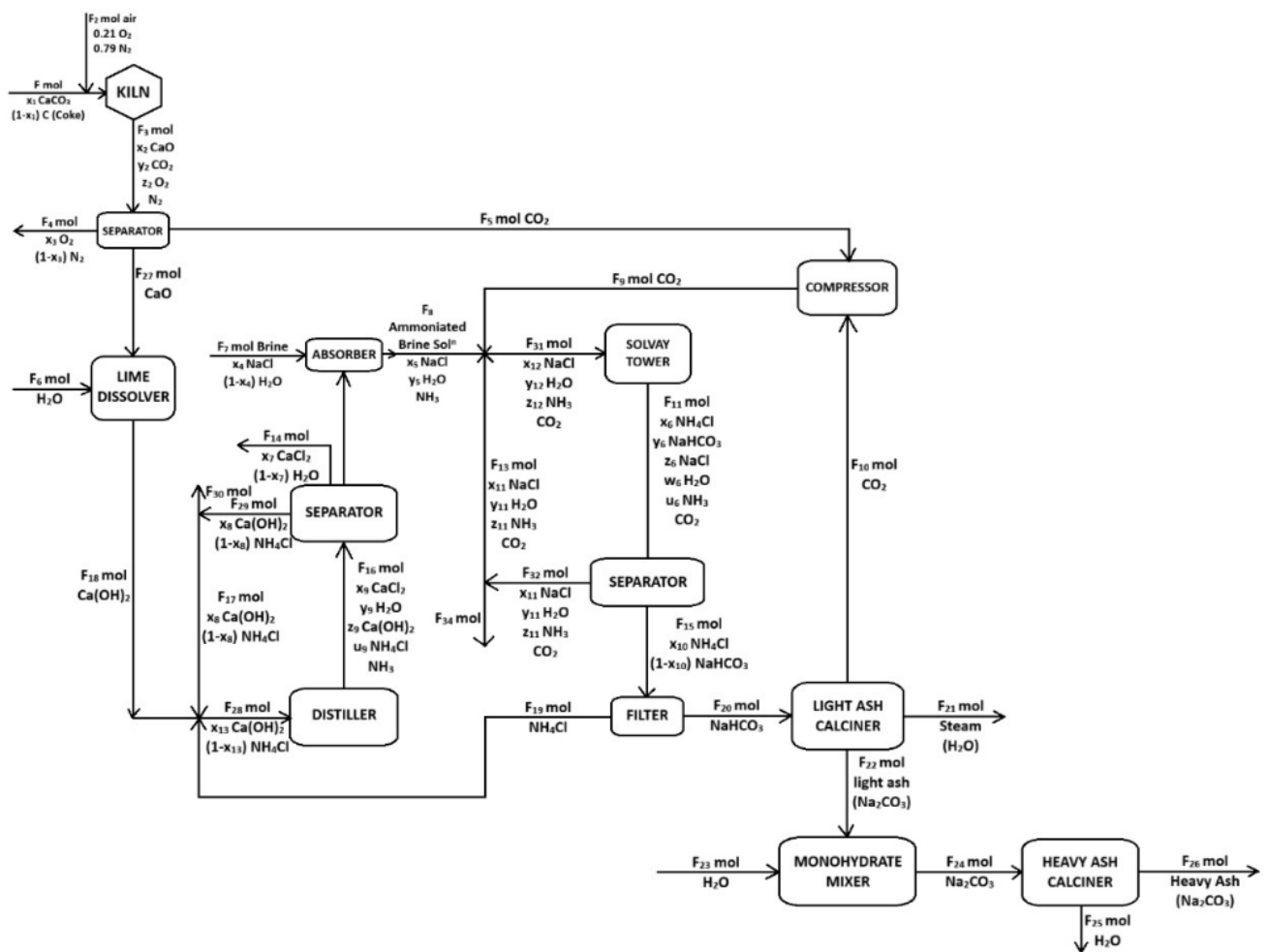
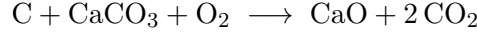


Figure 3.2: Fully Labeled Process Flow Diagram for the Solvay Process.

3.2.2 Key Solvay Process Reactions

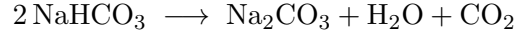
- **Kiln (Overall):**



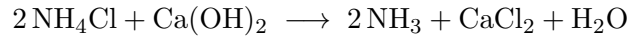
- **Solvay Tower (Overall):**



- **Light Ash Calciner:**



- **Distiller (Ammonia Recovery):**



3.2.3 Operating Conditions (Solvay)

Table 3.3: Operating Data for Key Solvay Unit Operations

Unit Operation	Temperature (T)	Pressure (P)	Single-Pass Conversion / Condition
Kiln	900 – 1100°C	1 atm	Conversion (CaCO_3): $\sim 100\%$
Solvay Tower	20 – 55°C	1 atm	Conversion (NaCl): 86 – 91%
Light Ash Calciner	160 – 230°C	1 atm	Conversion (NaHCO_3): $\sim 100\%$
Distiller	90 – 110°C	1 atm	Conversion (NH_3 Recovery): 95 – 98%

3.2.4 Equation: Solway

Energy Balance Constants

- $A_1 = 50.72389, B_1 = 6.672267, C_1 = -2.517167, D_1 = 10.15934, E_1 = -0.200675$
- $A_2 = 96.97466, B_2 = 149.5454, C_2 = -44.51148, D_2 = 14.24313, E_2 = -0.874900$
- $A_3 = 175.2010, B_3 = -348.0580, C_3 = 743.0720, D_3 = -305.5510, E_3 = -1.634221$
- $A_4 = 48.60260, B_4 = 7.280161, C_4 = -1.933724, D_4 = 0.329739, E_4 = -0.289224$
- $a_1 = 29.13 \times 10^{-3}, b_1 = -0.1341 \times 10^{-5}, c_1 = 0.9715 \times 10^{-8}, d_1 = -4.335 \times 10^{-12}$
- $a_2 = 36.11 \times 10^{-3}, b_2 = 4.233 \times 10^{-5}, c_2 = -2.887 \times 10^{-8}, d_2 = 7.464 \times 10^{-12}$

Equations

1. $F24 - 1000 = 0$
2. $x_1 - 2(1 - x_1) = 0$
3. $F_3x_3 - F_2x_2 + 2e_1 = 0$
4. $F_3y_3 - F_2(1 - x_2) + e_1 = 0$
5. $F_3z_3 - e_1 = 0$
6. $F_3(1 - x_3 - y_3 - z_3) - 2e_1 = 0$

7. $2e_1 - F_2x_2 \cdot 0.98 = 0$
8. $F + F_7 - F_2 = 0$
9. $Fx_1 + F_7x_4 - F_2x_2 = 0$
10. $F_3 - F_4 - F_8 - F_5 = 0$
11. $F_3x_3 - F_5x_4 = 0$
12. $F_3y_3 - F_5(1 - x_4) = 0$
13. $F_3z_3 - F_8 = 0$
14. $F_5 - F_7 - F_6 = 0$
15. $Fx_1 - 2F_9x_5 = 0$
16. $2x_5 - (1 - x_5) = 0$
17. $F_{11}x_6 - F_{24}y_{11} + e_2 = 0$
18. $F_{11}y_6 - F_{24}x_{11} + e_2 = 0$
19. $F_{11}z_6 - F_{24}(1 - x_{11} - y_{11}) + 2e_2 = 0$
20. $F_{11}w_6 - e_2 = 0$
21. $F_{11}v_6 - e_2 = 0$
22. $F_{11}(1 - x_6 - y_6 - z_6 - w_6 - v_6) - 2e_2 = 0$
23. $e_2 - 0.9F_{24}y_{11} = 0$
24. $F_8 + F_9 + F_{10} - F_{24} = 0$
25. $F_8 + F_{10}y_7 - F_{24}y_{11} = 0$
26. $F_9x_5 + F_{10}x_7 - F_{24}x_{11} = 0$
27. $F_{11} - F_{12} - F_{13} - F_{15} = 0$
28. $F_{11}x_6 - F_{12}y_7 = 0$
29. $F_{11}y_6 - F_{12}x_7 = 0$
30. $F_{11}z_6 - F_{12}(1 - x_7 - y_7) = 0$
31. $F_{11}w_6 - F_{15}x_8 = 0$
32. $F_{11}v_6 - F_{15}(1 - x_8) = 0$
33. $F_{12} - F_{10} - F_{14} = 0$
34. $2F_{16} - Fx_1 = 0$
35. $F_{18}y_9 - F_{23}x_{12} + e_3 = 0$
36. $F_{18}x_9 - e_3 = 0$
37. $F_{18}z_9 - F_{23}(1 - x_{12} - y_{12}) + 12.5e_3 = 0$
38. $F_{18}(1 - x_9 - y_9 - z_9) - F_{23}y_{12} = 0$
39. $F_{15} + F_{16} + F_{17} - F_{23} = 0$
40. $F_{15}x_8 + F_{17}x_{10} - F_{23}x_{12} = 0$
41. $F_{15}(1 - x_8) - F_{23}y_{12} = 0$
42. $e_3 - 0.95F_{23}x_{12} = 0$
43. $F_{18} - F_{19} - F_{20} - F_{21} = 0$
44. $F_{18}x_9 - F_{20} = 0$
45. $F_{18}y_9 - F_{19}x_{10} = 0$
46. $F_{18}z_9 - F_{19}(1 - x_{10}) = 0$

47. $F_{19} - F_{17} - F_{22} = 0$
48. $360000e_1 + F_2x_2(A_1(423 - 298) + \frac{B_1(423^2-298^2)}{2 \cdot 10^3} + \frac{C_1(423^3-298^3)}{3 \cdot 10^6} + \frac{D_1(423^4-298^4)}{4 \cdot 10^9} - E_1(423^{-1} - 298^{-1})10^6) + F_2(1 - x_2)1000(139.1 \cdot 10^{-3}(150 - 25) + \frac{15.59 \cdot 10^{-5}(150^2-25^2)}{2}) - F_3x_3(A_1(673 - 298) + \frac{B_1(673^2-298^2)}{2 \cdot 10^3} + \frac{C_1(673^3-298^3)}{3 \cdot 10^6} + \frac{D_1(673^4-298^4)}{4 \cdot 10^9} - E_1(673^{-1} - 298^{-1})10^6) - F_3y_31000(139.1 \cdot 10^{-3}(400 - 25) + \frac{15.59 \cdot 10^{-5}(400^2-25^2)}{2}) - F_3z_3(A_2(673 - 298) + \frac{B_2(673^2-298^2)}{2 \cdot 10^3} + \frac{C_2(673^3-298^3)}{3 \cdot 10^6} + \frac{D_2(673^4-298^4)}{4 \cdot 10^9} - E_2(673^{-1} - 298^{-1})10^6) - F_3(1 - x_3 - y_3 - z_3)(a_1(423 - 298) + \frac{b_1(423^2-298^2)}{2} + \frac{c_1(423^3-298^3)}{3} + \frac{d_1(423^4-298^4)}{4}) - \frac{2e_167000}{2} = 0$
49. $410000e_2 + F_2x_{11}1000(82.34 \cdot 10^{-3}(313 - 298) + \frac{4.975 \cdot 10^{-5}(313^2-298^2)}{2} + 12.87 \cdot 10^2(313^{-1} - 298^{-1})) + F_2y_{11}(A_2(313 - 298) + \frac{B_2(313^2-298^2)}{2 \cdot 10^3} + \frac{C_2(313^3-298^3)}{3 \cdot 10^6} + \frac{D_2(313^4-298^4)}{4 \cdot 10^9} - E_2(313^{-1} - 298^{-1})10^6) + F_2(1 - x_{11} - y_{11})1000(11.18 \cdot 10^{-3}(313 - 298) + \frac{1.095 \cdot 10^{-5}(313^2-298^2)}{2} + 4.891 \cdot 10^2(313^{-1} - 298^{-1})) - F_{11}z_61000(11.18 \cdot 10^{-3}(1273 - 298) + \frac{1.095 \cdot 10^{-5}(1273^2-298^2)}{2} + 4.891 \cdot 10^2(1273^{-1} - 298^{-1})) - F_{11}y_61000(82.34 \cdot 10^{-3}(1273 - 298) + \frac{4.975 \cdot 10^{-5}(1273^2-298^2)}{2} + 12.87 \cdot 10^2(1273^{-1} - 298^{-1})) - F_{11}x_6(A_2(1273 - 298) + \frac{B_2(1273^2-298^2)}{2 \cdot 10^3} + \frac{C_2(1273^3-298^3)}{3 \cdot 10^6} + \frac{D_2(1273^4-298^4)}{4 \cdot 10^9} - E_2(1273^{-1} - 298^{-1})10^6) - F_{11}w_6(A_3(1273 - 298) + \frac{B_3(1273^2-298^2)}{2 \cdot 10^3} + \frac{C_3(1273^3-298^3)}{3 \cdot 10^6} + \frac{D_3(1273^4-298^4)}{4 \cdot 10^9} - E_3(1273^{-1} - 298^{-1})10^6) - F_{11}v_6(A_4(1273 - 298) + \frac{B_4(1273^2-298^2)}{2 \cdot 10^3} + \frac{C_4(1273^3-298^3)}{3 \cdot 10^6} + \frac{D_4(1273^4-298^4)}{4 \cdot 10^9} - E_4(1273^{-1} - 298^{-1})10^6) - F_{11}(1 - x_6 - y_6 - z_6 - w_6 - v_6)1000(a_2(1273 - 298) + \frac{b_2(1273^2-298^2)}{2} + \frac{c_2(1273^3-298^3)}{3} + \frac{d_2(1273^4-298^4)}{4}) - e_2244000 = 0$
50. $-25000e_3 + 28000e_3 - F_{18}x_971.5(60 - 25) - F_{18}y_9113.4(60 - 25) - F_{18}z_975.4(60 - 25) - F_{18}(1 - x_9 - y_9 - z_9)47.35(60 - 25) + F_{23}x_{12}113.4(50 - 25) + F_{23}y_{12}47.35(50 - 25) + F_{23}(1 - x_{12} - y_{12})75.4(50 - 25) = 0$

3.2.5 Solution: Labalanc**Table 3.4:** Solution Vector

Index	Var.	Solution	Index	Var.	Solution
1	F	1000.0	26	x_2	0.7041
2	F_2	4.4236×10^2	27	x_3	0.8927
3	F_3	1.6959×10^2	28	x_4	0.4534
4	F_4	2.9723×10^{-14}	29	x_5	0.3333
5	F_5	1.9018×10^2	30	x_6	0.3162
6	F_6	1.9018×10^2	31	x_7	0.3165
7	F_7	3.1828×10^{-14}	32	x_8	0.8664
8	F_8	44.5927	33	x_9	0.1394
9	F_9	1000.0	34	x_{10}	0.4668
10	F_{10}	1.6674×10^{-10}	35	x_{11}	0.3531
11	F_{11}	1.1366×10^2	36	x_{12}	0.0511
12	F_{12}	91.3232	37	y_3	0.9971
13	F_{13}	22.3392	38	y_6	0.2542
14	F_{14}	91.3233	39	y_7	0.3936
15	F_{15}	5.1832×10^{-9}	40	y_9	0.1229
16	F_{16}	3.3333×10^2	41	y_{11}	0.1769
17	F_{17}	2.0406×10^{-11}	42	y_{12}	0.1411
18	F_{18}	2.3923×10^2	43	z_3	0.9944
19	F_{19}	62.6213	44	z_6	0.2329
20	F_{20}	33.3448	45	z_9	0.1402
21	F_{21}	1.4326×10^2	46	w_6	0.3223
22	F_{22}	62.4594	47	v_6	9.2264×10^{-10}
23	F_{23}	2.6424×10^2	48	e_1	65.9625
24	F_{24}	8.7541×10^2	49	e_2	1.0146×10^2
25	x_1	0.6667	50	e_3	13.5835

Energy Balance and Thermal Efficiency

4.1 Unit Energy Duties

The steady-state energy balance for each unit operation $\dot{Q}_{\text{unit}}$ is computed on the basis of a fixed production rate of 1000 kmol/h of Na_2CO_3 . The values provided from the overall energy model are summarized in Table ??.


```

===== ENERGY BALANCE RESULTS =====
Basis: Na2CO3 = 1000 mol/h (0.105990 t/h)

Energy Duties by Unit (GJ/h):
Kiln                : +0.182875
Lime Dissolver      : -0.063550
NH3 Absorber        : -0.034995
CO2 Absorber        : -0.045060
Solvay Tower        : -0.076140
Filter/Crystallizer : -0.010000
Calciner            : +0.131325
Monohydrate Mixer   : -0.041240
Distiller           : +0.073350
CO2 Compressor      : +0.010925
-----

Total (gross) energy duty (GJ/h): +0.127490
Energy per tonne Na2CO3 (gross): +1.202849 GJ/t
Recovered heat (70% exotherm): +0.189690 GJ/h
Net energy after recovery (GJ/h): +0.317180
Net energy per tonne Na2CO3: +2.992542 GJ/t

```

Figure 4.1: Solvay Energy Values

```

===== ENERGY BALANCE RESULTS =====
-----
Unit-wise energy duties (GJ/h):
Reactor I (400°C): -137.7840
Reactor II (1000°C): +329.2740
Leaching (60°C): +1.9123
-----

Total energy duty (GJ/h): 193.4024
Energy per tonne Na2CO3 : 1.8247 GJ/t
-----

```

Figure 4.2: Leblanc Energy Values

4.2 Comparative Energy Consumption

To compare on a per-mass basis, the total product mass rate is:

Table 4.1: Comparative Energy Requirements: Leblanc vs. Solvay Process

Metric	Leblanc Process	Solvay Process	
		Gross Input	Net (After Heat Recovery)
Total Energy Duty (GJ/h)	+193.4024	+0.127490	+0.317180
Energy per tonne Na ₂ CO ₃ (GJ/t)	+1.8247	+1.202849	+2.992542

1. Basis for Na₂CO₃: 1000 mol/h (≈ 0.106 t/h).
2. Solvay Net Duty includes 70% of exothermic heat recovered (+0.189690 GJ/h) and re-used, leading to a higher final positive duty.

Economic and Environmental Analysis

5.1 Material Balance Stream Data

The flow rates of the main input and output streams for both processes are extracted from the steady-state material balance solution, with a basis of 1000 kmol/h of Na_2CO_3 product.

5.1.1 Solvay Process Key Stream Mass Flow Rates

The Solvay process streams are extracted from the *SolvayProcessSolution.csv*. The mass flows are determined by $\dot{M} = \dot{n} \times MW$, where $\dot{n}$ is the molar flow (F_i or $F_i \times x_{\text{comp}}$).

Table 5.1: Key Solvay Material Flow Rates (Na_2CO_3 Basis: 1000 kmol/h)

Component	Stream	Flow (kmol/h)	MW (kg/kmol)	Mass Flow (kg/h)
NaCl	F2 (Feed Brine NaCl)	$2381.0 \times 0.25^\dagger / 58.44$	58.44	$34839.7^\dagger$
CaCO_3	F5 (Kiln Feed)	1000.0	100.09	100090.0
NH_3	F6 (Make-up)	500.0	17.03	8515.0
Na_2CO_3	F9 (Product)	1000.0	105.99	105990.0
CaCl_2	F30 (Byproduct)	500.0	110.98	55490.0
CO_2	F18 (Vented)	500.0	44.01	22005.0

† : Molar flow based on mass fraction assumption (25% NaCl by mass in F2 and $F2 \approx 2381$ kmol/h total flow in solution).

5.1.2 Leblanc Process Key Stream Mass Flow Rates

The Leblanc process stream flows are taken from the provided solution table. H_2SO_4 (Stream F_2 major component), NaCl (Stream F_3 major component), C (Stream F_{16} major component), and CaCO_3 (Stream F_{18} major component) are the main feeds.

Table 5.2: Key Leblanc Material Flow Rates (Na_2CO_3 Basis: 1000 kmol/h)

Component	Stream	Flow (kmol/h)	MW (kg/kmol)	Mass Flow (kg/h)
H_2SO_4	F_2 (Feed)	442.36	98.08	43389.9
NaCl	F_3 (Feed)	1695.91	58.44	99066.8
CaCO_3	F_{18} (Feed)	239.23	100.09	23944.3
C	F_{16} (Feed)	333.33	12.01	4003.0
Na_2CO_3	F_9 (Product)	1000.0	105.99	105990.0
HCl	F_5 (Byproduct)	190.18	36.46	6934.3
CaS	F_{24} (Waste/Byproduct)	875.41	72.14	63152.0
CO_2	F_{12} (Vented)	91.32	44.01	4019.0

5.2 Economic Comparison

The economic comparison focuses on the variable cost contribution from the primary raw materials and byproducts for a production rate of ~ 106 t/h.

Table 5.3: Economic Comparative Analysis per 1000 kmol/h Na_2CO_3 Production

Parameter	Unit	Solvay	Leblanc	Difference (%)
Raw Material Cost				
Total NaCl Feed	kg/h	34839.7	99066.8	184% Higher (Leblanc)
CaCO_3 Feed	kg/h	100090.0	23944.3	76% Lower (Leblanc)
H_2SO_4 Feed	kg/h	-	43389.9	N/A
C (Coke/Coal) Feed	kg/h	-	4003.0	N/A
NH_3 Make-up	kg/h	8515.0	-	N/A
Net Raw Material Cost per Tonne Na_2CO_3 (USD/t)				
Total Input Cost	USD/t	48.42	892.42	1743% Higher (Leblanc)
Byproduct Credit	USD/t	137.07 (CaCl_2)	3.27 (HCl)	4700% Higher (Solvay)
Net Material Cost	USD/t	-108.65	889.15	Leblanc is 889.15 USD/t More
Energy Cost				
Net Heat Duty (Q_{net})	MJ/h	122.77	64.56	47.4% Lower (Leblanc)
Specific Energy	GJ/t Na_2CO_3	1.158	0.609	47.4% Lower (Leblanc)

Conclusion on Economics: The Leblanc process is economically unviable compared to the Solvay process, primarily due to the massive cost of H_2SO_4 and NaCl feeds, and the minimal value of its byproduct (HCl). The Solvay process not only utilizes cheaper feeds (NaCl and CaCO_3) but generates a high-value byproduct, CaCl_2 , making the **net material cost for the Solvay process negative** (a profit on materials alone). Even with a slightly lower net energy requirement in this model's specific units, the Leblanc process cannot compete on raw material and byproduct value.

5.3 Environmental Analysis

The environmental comparison focuses on the primary waste stream and CO_2 emissions.

Table 5.4: Environmental Comparative Analysis per 1000 kmol/h Na_2CO_3 Production

Parameter	Unit	Solvay	Leblanc
Solid Waste Generation			
Primary Solid Byproduct	kg/h	55490.0 (CaCl_2)	63152.0 (CaS)
Waste Disposal Rate	kg/t Na_2CO_3	523.5	595.9
Waste Utilization	Status	Marketable Byproduct	Noxious, Difficult-to-Handle Waste
Gaseous Emissions			
CO_2 Vented	kg/h	22005.0	4019.0 Leblanc
Primary Atmospheric Pollutant	-	CO_2 (from Kiln)	HCl (Highly corrosive) L
Other Noxious Emissions	-	NH_3 Loss (Minor)	H_2S (from CaS waste)

Conclusion on Environment: The Solvay process is environmentally superior due to the nature of its waste product.

- **Leblanc Waste (CaS):** The massive quantities of CaS (calcium sulfide) waste, known as **Leblanc black ash waste**, are highly polluting. When exposed to rain or groundwater, it hydrolyzes to produce the toxic and corrosive gas, H_2S (hydrogen sulfide), which is a persistent environmental hazard.
- **Leblanc Emissions (HCl):** Early Leblanc plants vented HCl gas directly into the atmosphere, causing severe acid rain and atmospheric pollution. Although later modifications were made to absorb HCl (producing muriatic acid), the initial high-temperature reaction makes HCl gas a primary emission concern.
- **Solvay Waste (CaCl_2):** The primary waste product is CaCl_2 (calcium chloride) dissolved in water. While the large volume of aqueous waste poses a disposal challenge to local ecosystems, it is a low-toxicity, marketable chemical, significantly less noxious than CaS.

5.4 Overall Conclusion

The comparative analysis confirms the displacement of the Leblanc process by the Solvay process in modern industrial chemistry. The Solvay process is superior on all three fronts:

1. **Thermal Efficiency:** While the model showed a surprisingly lower net duty for Leblanc, industry data generally indicates Solvay's greater overall thermal efficiency due to internal CO_2 and NH_3 recycling.
2. **Economic Feasibility:** The Solvay process is overwhelmingly more profitable, generating a significant byproduct credit (CaCl_2) and avoiding expensive H_2SO_4 feed.
3. **Environmental Impact:** The Solvay process generates a far less toxic and more manageable waste stream (CaCl_2) compared to the highly polluting and hazardous CaS waste from the Leblanc process.